

PRODUCE COLD STORAGE & POSTHARVEST HANDLING TECHNICAL MANUAL

Comprehensive Commodity Preservation Standards

EXECUTIVE SUMMARY & SCOPE: Postharvest quality retention and thermodynamic stability depend on matching commodity intrinsic matrix properties (water activity a_w , moisture content, lipid concentration, pH, and respiratory kinetics) with extrinsic environmental storage parameters (temperature, relative humidity [RH], ethylene dynamics, and modified atmosphere containment). This technical manual details operational protocols and failure mechanisms across four major commodity classes: Active Horticultural Tissues, Dairy Solids & Emulsions, Low-Moisture Baked Commodities, and Fermented & Acidified Bio-Systems.

1. Active Horticultural Tissues (Fruits & Vegetables)

Fresh fruits and vegetables remain living, metabolically active tissues postharvest. Cellular respiration oxidizes stored starches, sugars, and organic acids into CO_2 , water vapor, and metabolic heat. Postharvest losses typically range from 10% to 40% when environmental controls fail to suppress metabolic and transpiration rates.

Respiration Rates & Metabolic Heat Dynamics

Respiration rates dictate commodity perishability. Crops with extremely high respiration rates ($>60 \text{ mL CO}_2/\text{kg}\cdot\text{hr}$ or $>60,000 \text{ BTU/ton/24 hr}$) such as asparagus, broccoli, sweet corn, mushrooms, and spinach require immediate precooling to prevent rapid sugar depletion, tissue yellowing, and thermal self-heating. Conversely, low-respiration commodities (apples, garlic, onions, mature potatoes) exhibit extended cold storage potential.

Temperature, Relative Humidity & Transpiration Control

Cool-season crops (leafy greens, root crops, brassicas, apples) thrive near freezing ($32\text{--}35^\circ\text{F}$ / $0\text{--}2^\circ\text{C}$) with 90–100% RH to minimize the vapor pressure deficit (VPD) and prevent desiccation, wilting, and weight loss. High RH is maintained using air humidifiers, wetting storage floors, package icing, or micro-perforated film liners. Refrigeration coils must be kept within 2°F of storage room air temperature to prevent rapid moisture desiccation.

Chilling Injury vs. Freezing Injury Mechanisms

Chilling Injury (CI): Warm-season commodities (cucumbers, eggplants, sweet peppers, tomatoes, melons, sweet potatoes) suffer physiological breakdown when stored below critical thresholds ($36\text{--}55^\circ\text{F}$ / $2\text{--}13^\circ\text{C}$). Sub-threshold temperatures disrupt membrane lipid fluidity, causing membrane phase transitions from liquid-crystalline to solid-gel states. Symptoms include surface pitting, water-soaked lesions, internal tissue browning, seed darkening, failure to ripen, and heightened vulnerability to opportunistic pathogens (*Alternaria*, *Botrytis*).

Freezing Injury: Occurs below the commodity freezing point ($<28.8\text{--}31.7^\circ\text{F}$). Ice crystal formation ruptures cell walls, leading to loss of turgor rigidity, water-soaking, and complete collapse upon thawing.

Commercial Precooling Methodologies

- **Hydrocooling:** Rapid heat removal using chilled water (32–35°F). Essential for asparagus, sweet corn, and celery. Requires sanitation (chlorination) to prevent pathogen distribution.
- **Forced-Air Cooling:** Pulls cold air through packed containers. Ideal for berries, peppers, and packaged snap beans.
- **Vacuum Cooling & Hydrovacuum:** Evaporaporative cooling of leafy greens (head lettuce) by pulling a vacuum; water evaporates from tissue surfaces, dropping core temperatures to 32°F in 20–30 minutes.
- **Package Icing / Liquid Ice:** Injected into waxed cartons for broccoli and green onions to maintain 100% RH.

Storage Compatibility Classification (Groups 1–7)

Group	Temp (°F / °C)	RH (%)	Ethylene Profile	Representative Commodities
Group 1	32–36°F (0–2°C)	90–95%	Ethylene Producers	Apples, Pears, Peaches, Plums, Cherries, Berries, Topped Beets, Mushrooms
Group 2	32–36°F (0–2°C)	95–100%	Ethylene Sensitive	Asparagus, Broccoli, Cabbage, Carrots, Cauliflower, Celery, Leafy Greens, Radishes
Group 3	32–36°F (0–2°C)	65–75%	Moisture Sensitive	Dry Garlic, Dry Onions (High RH induces root growth & decay)
Group 4	40–45°F (4.5–7°C)	90–95%	CI Sensitive / Odor Risk	Cantaloupes, Cranberries, Lemons, Limes, Oranges, Tangerines, Yucca Root
Group 5	50°F (10°C)	85–90%	Ethylene & CI Sensitive	Cucumbers, Eggplants, Okra, Sweet Peppers, Storage Potatoes, Summer Squash
Group 6	55–60°F (13–15°C)	85–90%	Ethylene Producers & CI	Avocados, Bananas, Grapefruit, Guava, Mangoes, Melons, Papayas, Ripe Tomatoes
Group 7	65–70°F (18–21°C)	85–90%	Ripening / CI Sensitive	Ginger, Jicama, Sweet Potatoes, Watermelon, Yams, Mature Green Tomatoes

2. Dairy Commodity Class & Matrix Stability

Dairy preservation spans low-moisture powders, high-fat emulsions, and cultured protein gel networks. Stability depends on controlling lactose state transitions, lipid autooxidation, lipolysis, casein protein coagulation, and syneresis.

Lactose State Transitions, Caking & Maillard Browning

Dehydrated dairy powders (Whole Milk Powder [WMP] and Skim Milk Powder [SMP]) contain lactose in a metastable, amorphous glassy state. If ambient storage RH exceeds 65% or powder moisture rises above 4.0%, water sorption depresses the glass transition temperature (T_g) below storage temperature. This triggers an amorphous-to-crystalline phase transition into α -lactose monohydrate. Crystallization expels bound water into the matrix, causing particle clumping, severe caking, loss of solubility, and rapid Maillard browning between lactose reducing sugars and lysine amino acid residues.

Lipid Autooxidation & Hydrolytic Lipolysis in Dairy Fats

WMP contains $\geq 26.0\%$ milkfat, whereas SMP contains $\leq 1.5\%$ fat. Free surface lipids on WMP particles act as primary sites for autooxidation, reacting with dissolved oxygen to form free radicals, hydroperoxides, and volatile off-flavor aldehydes (hexanal). WMP requires nitrogen gas flushing or vacuum sealing in light-barrier packaging ($<25^\circ\text{C}$, $<65\%$ RH) to achieve a 12–24 month shelf life. In high-moisture/high-fat matrices (butter, cheddar cheese), active or residual heat-resistant lipases cleave triacylglycerols into short-chain free fatty acids (butyric, caproic acids), inducing hydrolytic rancidity.

Cultured Dairy Fermentation, Casein Coagulation & Syneresis

Yogurt and fermented dairy products utilize *Streptococcus thermophilus* and *Lactobacillus delbrueckii* subsp. *bulgaricus* to ferment lactose into lactic acid. Heating milk to 165–200°F (74–93°C) prior to inoculation denatures whey proteins (β -lactoglobulin), enabling them to complex with κ -casein. As acid production drops the pH to the casein isoelectric point (pH 4.6), casein micelles aggregate into a three-dimensional protein gel network.

Syneresis & Post-Acidification: Storage at 32–39°F (0–4°C) halts continued LAB acid production. Over-incubation or temperature abuse causes post-acidification (pH <3.8), leading to protein network contraction and syneresis (whey separation). Stabilizers (pectin, low acyl gellan gum, transglutaminase enzymatic cross-linking) are applied to reinforce gel firmness.

Dairy Commodity	Moisture (%)	Fat Content (%)	pH Range	Primary Failure Mechanism	Storage Protocol
Whole Milk Powder	2.0–4.0%	≥26.0%	6.5–6.7 (rec)	Amorphous lactose caking & lipid autoxidation	<77°F (<25°C), <65% RH, N2-flushed barrier bags
Skim Milk Powder	3.0–4.5%	≤1.5%	6.5–6.8 (rec)	Lactose crystallization & Maillard browning	<77°F (<25°C), <65% RH, multi-ply paper/PE liner
Cheddar Cheese	36.0–39.0%	30.0–34.0%	5.10–5.35	Pathogen growth, surface mold, lipolytic rancidity	32–42°F (0–5.5°C), 80–85% RH, vacuum shrink-wrap
Plain Yogurt	85.0–88.0%	3.25–4.0%	3.80–4.65	Post-acidification, syneresis (whey separation)	32–39°F (0–4°C), sealed PP cups with foil lids
Salted Butter	15.5–16.5%	80.0–82.0%	6.1–6.4	Hydrolytic lipolysis & light-induced oxidation	32–39°F (0–4°C), foil laminate wrapper, <75% RH

3. Low-Moisture & Baked Commodity Classes

In dry snacks, crackers, crisps, and bakery goods, preservation centers on controlling water activity (a_w) and managing the thermodynamic state of starch/protein matrices and lipid fractions.

Water Activity (a_w) Dynamics & Sorption Isotherms

Water activity measures the energetic availability of water in a food matrix, defined as the vapor pressure of food water divided by pure water vapor pressure at the same temperature ($a_w = p / p_0 = ERH / 100$). Unlike total moisture content, a_w dictates chemical reactivity, enzymatic rates, and microbial growth limits.

Critical Water Activity Thresholds

- **$a_w < 0.60$:** Absolute limit for all microbial growth. Bacteria stop at a_w 0.87–0.91; molds and yeasts stop at a_w 0.70–0.80.
- **$a_w = 0.35$ – 0.45 (Crispness Loss & Glass Transition):** Crisps, crackers, and puffed snacks exist in a rigid, glassy structural state. Moisture sorption raises internal a_w above critical thresholds ($a_w > 0.35$ – 0.45), depressing the glass transition temperature (T_g) below ambient storage. Water acts as a plasticizer, increasing molecular mobility and causing a phase transition into a rubbery, soft state.
- **Multi-Component Moisture Migration:** In filled pastries or snack cakes (e.g., cream-filled cakes), water migrates from regions of high a_w (filling, a_w 0.80) to low a_w (crust/cookie, a_w 0.30). Components must be formulated to identical a_w targets to eliminate the thermodynamic driving force for migration.

Lipid Oxidation Kinetics in Low-Moisture Foods

In deep-fried potato crisps (25–38% fat) and crackers (8–20% fat), lipid oxidation exhibits a parabolic relationship with water activity. At extremely low water activity ($a_w < 0.10\text{--}0.20$), lipid oxidation accelerates significantly because un-hydrated catalysts (trace metals) directly contact fat molecules and protective hydrogen-bonded water films are absent. Furthermore, confocal microscopy demonstrates that baked crackers possess highly porous structures; surface lipids and interior lipids oxidize at identical rates because oxygen diffuses freely throughout the porous matrix.

Barrier Packaging Specifications

- **Water Vapor Transmission Rate (WVTR):** Moisture-sensitive crisp snacks require high WVTR barriers (metallized oriented polypropylene [mOPP] / aluminum foil laminates) to prevent environmental water absorption.
- **Oxygen Transmission Rate (OTR) & MAP:** High-fat snacks require oxygen-impermeable barriers combined with modified atmosphere packaging (high-purity N₂ flushing) to reduce headspace O₂ to <1.0%, halting free radical initiation.

Baked / Snack Item	Moisture (%)	Fat (%)	Water Activity (a_w)	Primary Spoilage / Failure	Packaging Barrier Need
Potato Crisps	1.2–2.5%	30.0–38.0%	0.15–0.25	Loss of crispness ($a_w > 0.35$) & oxidation	N2-flushed aluminum foil pouch (High WVTR & OTR)
Puffed Corn Snacks	1.5–3.0%	22.0–32.0%	0.18–0.28	Textural softening & rancidity	High-barrier foil flow-pack under MAP
Wheat Crackers	2.0–4.0%	8.0–20.0%	0.20–0.30	Starch matrix rubberiness & hexanal off-flavor	Waxed inner wrap in paperboard/foil laminate
Soft Cookies	8.0–12.0%	10.0–18.0%	0.60–0.75	Staling, mold growth, moisture loss	Hermetic moisture-retentive polymer film
Fresh Bread / Cakes	35.0–45.0%	2.0–8.0%	0.88–0.95	Mold growth (Penicillium) & staling	Polyethylene bags with preservatives / MAP

4. Fermented & Acidified Commodities

Fermented vegetables (Kimchi, Sauerkraut, Pickles) rely on acid suppression, osmotic stress, and anaerobic conditions to prevent microbial spoilage and achieve long-term biochemical stability.

Lactic Acid Bacteria (LAB) Succession Kinetics

Spontaneous vegetable fermentations follow a strict ecological succession of wild lactic acid bacteria:

1. **Initiation Phase (Heterofermentative):** *Leuconostoc mesenteroides* dominates early fermentation at 18–22°C (65–72°F). It metabolizes vegetable sugars into D-lactic acid, acetic acid, CO₂, ethanol, and mannitol. Carbonation creates an anaerobic environment while lowering the pH from 6.0 to ~4.5.
2. **Acidification Phase (Homofermentative):** As acidity accumulates, acid-tolerant *Lactobacillus plantarum* and *Lactobacillus brevis* outcompete *Leuconostoc*, converting remaining hexose sugars purely into L-lactic acid, driving the pH down to its final equilibrium of 3.3–4.2.

Brine Osmosis & Critical pH Safety Thresholds

- **Salting & Osmosis (2.0–5.0% NaCl):** Dry salting or brine soaking draws out cell sap water and soluble nutrients through osmotic pressure. Salt inhibits spoilage gram-negative bacteria (*Enterobacteriaceae*) while selecting for salt-tolerant LAB.
- **Botulism & Pathogen Suppression (pH < 4.6 & pH < 4.0):** The strict regulatory safety threshold for acidified/fermented foods is **pH < 4.6**, which completely inhibits spore germination and toxin production by *Clostridium botulinum*. Optimal fermentation reaches **pH 3.3–4.0**, where non-dissociated organic acids penetrate microbial cell membranes, ionize within the neutral cytoplasm, collapse the transmembrane proton gradient, and kill pathogenic strains (*E. coli*, *Listeria*).

Post-Fermentation Chilling & Texture Retention

- **Temperature Control (-1 to 4°C / 30 to 39°F):** Once optimal titratable acidity (0.6–0.9% lactic acid) and pH (3.8–4.2) are achieved, ferments must be immediately chilled to -1 to 4°C. Chilling halts LAB metabolic kinetics, preventing over-acidification (>0.9% acid), sour off-flavors, and tissue softening.
- **Pectin Structural Retention:** Enzymatic degradation of cell wall pectins by polygalacturonase leads to mushy textures. Addition of divalent calcium ions (0.1–0.5% CaCl₂) cross-links polygalacturonic acid chains, forming insoluble calcium pectate gels that maintain vegetable crispness.

Fermented Product	Salt Brine (%)	pH Threshold	Dominant Microflora	Quality Failure Mode	Chilled Storage Protocol
Cabbage Kimchi	1.5–4.0% NaCl	pH 3.8–4.2 (Opt)	Leuconostoc, Lactobacillus, Weissella	Over-acidification (>0.9% acid), softening, gas pressure	30–39°F (-1 to 4°C), gas-vented containers, 3–12 mos
Sauerkraut	2.0–2.5% NaCl	pH 3.3–3.8	L. mesenteroides -> L. plantarum	Kahm yeast film, softening, putrid odor if un-submerged	32–39°F (0–4°C), submerged in brine under weight, 6+ mos
Fermented Dill Pickles	3.0–5.0% NaCl	pH 3.4–3.8	Lactobacillus plantarum, Pediococcus	Hollow pickles, softening (pectin loss), bloater damage	34–40°F (1 to 4°C) or hot-filled pasteurized glass
Pickled Jalapeños	Direct Acid + Salt	pH 3.1–3.6	Acidified (Acetic Acid / Vinegar)	Softening, loss of capsaicin pungency, ring clouding	Ambient room temp (59–77°F) if hot-filled & sealed

5. Integrated Cross-Commodity Preservation Matrix

Comprehensive food supply chain management requires applying distinct thermodynamic and biological control regimes tailored to each specific commodity matrix:

1. **Living Horticultural Regimes:** Requires respiratory kinetic suppression via strict cold chain management (32–55°F), high relative humidity (85–100%) to arrest transpiration, and ethylene segregation.
2. **Dehydrated Glassy Matrix Regimes:** Requires maintaining water activity below critical glass transition thresholds ($a_w < 0.35–0.45$ for snacks; $a_w < 0.20$ for powders) to prevent structural softening, caking, and lactose crystallization.
3. **High-Fat & Emulsion Regimes:** Requires atmospheric oxygen control (N₂ MAP flushing, OTR barriers) and light exclusion to stop radical-mediated autoxidation and hydrolytic rancidity.
4. **Fermented Bio-System Regimes:** Requires controlled LAB acid production (pH < 4.6), osmotic brining, and prompt post-fermentation chilling (-1 to 4°C) to lock in acidity and structural crispness.